

Suggested FYP topics for 2026-27

by

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Project code :	DWX1
Project title :	Finding ambiguities in network protocol standards through cross-implementation analysis
No. of students :	2
Description of project :	Network protocol standards (RFCs) define how systems on the Internet communicate. But these standards sometimes leave important behaviors unspecified: they simply never say what an implementation should do in certain situations. When this happens, different implementations of the same protocol must invent their own behavior, and they often arrive at different answers. These silent disagreements have real security consequences: they enable fingerprinting attacks, active probing of encrypted services, and parser differential exploits. In this project, students will develop a methodology to systematically discover these hidden ambiguities. The approach works by analyzing the source code of multiple open-source implementations of the same protocol using language models (LLM), extracting behavioral rules that describe how each implementation handles protocol events, and then comparing these rules across implementations. In other words, we will write “RFCs” backwards from implementations. Where implementations disagree, students will cross-reference against the original RFC to determine whether the standard specifies the behavior or is silent. Cases where the RFC is silent represent genuine under-specifications, which are gaps in the standard with real-world consequences. Students will apply this methodology to one or more protocols such as TLS, DNS, or SIP, and evaluate the security impact of any discovered ambiguities. Students are expected to have sufficient knowledge of computer networking and LLM. Program analysis skill is a plus.
Industrial collaboration :	No.

Project code :	DWX2
Project title :	Fingerprinting the Fingerprintless Browsers by their Instability Patterns
No. of students :	2
Description of project :	Anti-detect browsers (e.g., Multilogin, GoLogin, AdsPower) are modified Chromium browsers that generate isolated browsing profiles with spoofed fingerprints, enabling operators to run hundreds of apparently distinct "users" from a single machine. They are widely used in e-commerce multi-account management, ad fraud, and credential abuse, yet have received almost no academic study. In this project, students will investigate whether these so-called “fingerprintless” browsers can be effectively fingerprinted by analyzing how their browser-fingerprints change over time. Real browsers exhibit characteristic fingerprint instability patterns rooted in hardware and OS interactions (e.g., GPU rendering variations, font subsystem timing, OS window management). Anti-detect browsers must either suppress this

	natural instability or simulate it unconvincingly: both strategies are detectable. Students will build a web-based fingerprinting test page, collect repeated measurements from multiple anti-detect browsers and real browsers, and develop a classifier that distinguishes real devices from spoofed profiles. Students are expected to have sufficient knowledge of computer networking. Past experience with web development is a plus.
Industrial collaboration :	No.

Project code :	DWX3
Project title :	Is Email Address Munging Still Effective?
No. of students :	1-2
Description of project :	Email address munging—disguising email addresses on web pages (e.g., writing "user AT example DOT com")—is a widespread practice to prevent automated harvesting by spammers. Prior studies showed that even simple obfuscation defeated the regex-based scrapers of their era, but modern large language models can semantically parse obfuscated text that old scrapers could not. In this project, students will conduct a measurement study with three components: (1) scrape academic department websites to build a large-scale taxonomy of munging techniques in the wild, (2) build a pipeline combining regex, LLMs, and OCR to test how many munged addresses can be automatically extracted, and (3) analyze email address predictability from organizational naming patterns (e.g., firstname.lastname@university.edu). Students should be familiar with web scraping. Experience with LLM APIs is helpful.
Industrial collaboration :	No.